Product Requirements Document (PRD)

Product Name: Duplicate Bin

Document Code: PRD-DB-2026-V1.0

Product Lead / Owner: Senior Product Analyst / Technical BA

Target Release: Version 0.2.0 Desktop + Version 1.0.0 SaaS Relay

Status: Approved Baseline

1. Product Vision & Value Proposition

Product Vision:

"To become the world's most trusted, performant, and privacy-respecting storage optimization ecosystem, empowering individuals and distributed enterprise teams to effortlessly reclaim digital storage, slash recurring cloud costs, and govern unstructured data without risk of data loss."

Value Proposition:

Duplicate Bin solves digital storage clutter by combining ultra-fast local cryptographic detection with the oversight of a centralized cloud management platform. Unlike legacy utilities that risk permanent file deletion, Duplicate Bin introduces a guaranteed reversible quarantine system and 8 automated selection heuristics, turning hours of agonizing file reviews into a 10-second background operation.

2. Target Personas & User Journey Maps

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│ PERSONA 1: Elena - The Commercial Photographer (Prosumer)            │
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│ • Demographics: 29, Freelance Media Creator, 8 TB Local NVMe + Google Drive │
│ • Core Pain: Storage limit warnings, duplicates scattered across 20 folders │
│ • Goal: Fast visual identification of duplicate shots; keep RAW, trash JPEG │
│ • Favorite Feature: Drag & Drop Ingestion + "Keep Best Quality" Strategy    │
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│ PERSONA 2: Marcus - The Digital Agency IT Lead (Enterprise B2B)      │
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│ • Demographics: 41, Systems Administrator managing 35 creative workstations │
│ • Core Pain: Shared office NAS filling up; employees hoarding duplicates    │
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• Goal: Centralized visibility; automated remote scans without user friction
• Favorite Feature: 6-Digit Device Pairing + Cloud SaaS Dashboard Telemetry
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3. Product Goals & Non-Goals

Product Goals (What We Are Building)

1. **Unrivaled Scan Velocity:** Scan 10,000 files in < 20 seconds, maintaining 3-5x faster rescans through persistent SQLite hash caching.
2. **Absolute Data Protection:** 100% rollback capability via non-destructive quarantine archiving.
3. **Frictionless Onboarding:** Sub-60-second time-to-first-scan with zero-configuration drag-and-drop.
4. **Hybrid Monetization Engine:** Profitable, low-friction pay-as-you-go credit model ($4.99/100GB) and enterprise team tiers ($29 - $99/mo).

Non-Goals (Explicitly Out of Scope for Current Cycle)

- We are **not** building a cloud backup/storage provider (we optimize existing storage, not host files).
- We are **not** building a mobile photo gallery app (focus remains on desktop and networked NAS storage).
- We are **not** implementing destructive permanent file wiping by default (safety quarantine is mandatory).

4. Feature Prioritization Framework (MoSCoW)

Every feature in the product backlog is prioritized with explicit business and engineering rationale:

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│ MUST HAVE (Mandatory for Product Viability)                          │
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│ • Multi-Algorithm Cryptographic Hashing (SHA-256, BLAKE2b, MD5)      │
│   Rationale: Absolute mathematical proof of duplicate identity is non-negotiable.
│ • Two-Pass Size Pre-Filter Engine                                    │
│   Rationale: Prevents system crashing and scales scanning to 100k+ files.
│ • Safe Archive Quarantine & 1-Click Restore                          │
│   Rationale: Overcomes the primary psychological barrier to adoption: fear.
│ • Core Selection Strategies (Keep Newest, Oldest, Best Quality)      │
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  Rationale: Eradicates manual checkbox fatigue.
• Device Pairing & Telemetry Ingestion (SaaS Backend)
  Rationale: Foundational pipeline for corporate SaaS monetization.
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SHOULD HAVE (High-Value User Experience Enhancements)
• Drag-and-Drop Folder Ingestion Widget
  Rationale: Modernizes desktop interaction; reduces cognitive friction.
• Persistent SQLite WAL Hash Cache
  Rationale: Drives 80% latency reduction on repeat scans.
• Side-by-Side Visual Previews (Images, Audio, PDF)
  Rationale: Provides reassuring visual proof before user confirms cleanup.
• Centralized SaaS Storage Savings Dashboard
  Rationale: Gives IT buyers proof of ROI to justify subscription renewal.
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COULD HAVE (Delighters & Power-User Features)
• 13 Global Keyboard Shortcuts (Ctrl+O, Ctrl+S, F1, etc.)
  Rationale: Appeals to developer and power-user prosumer workflows.
• Modular Plugin Architecture (Desktop Notifications, Stats Logger)
  Rationale: Fosters community open-source extensions.
• Multi-Language Support (English, Spanish, French, German, Chinese, Japanese)
  Rationale: Expands international distribution reach.
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WON'T HAVE (Deferred to Future Milestones)
• Direct Google Drive / Dropbox Remote API Scanning
  Rationale: Requires extensive OAuth token maintenance and cloud rate-limit
  handling; users currently achieve this via local virtual sync folders.
• Perceptual Hashing (pHash) for Resized Images
  Rationale: Requires complex AI image processing dependencies (`imagehash`).
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5. Success Metrics & Product KPIs

Objective	Metric Indicator	Baseline	Target (v0.2.0 / v1.0.0)
Performance	Initial Scan Time (10k files)	60+ seconds (Competitors)	**18.7 seconds**
Efficiency	Rescan Time (10k files cached)	45 seconds	**4.2 seconds**

Objective	Metric Indicator	Baseline	Target (v0.2.0 / v1.0.0)
Safety	User Support Tickets for Data Loss	Industry average 0.5%	**0.0%** (Safe Archive)
Monetization	Credit Pack Conversion Rate	N/A (New release)	**12%** of active users
Engagement	Average Storage Reclaimed per Scan	0 MB	**15 - 35 GB**

6. Release & Go-To-Market Strategy

Phase 1: Community Desktop Open-Source Launch (v0.2.0)

- Release native packages: Debian repository `.deb` package (187 KB), Windows standalone `.exe`, and macOS universal binary.
- Position as a fast, beautiful, privacy-first desktop utility to build creator goodwill and gather bug telemetry.

Phase 2: Hybrid SaaS Beta (v1.0.0)

- Soft launch web portal and FastAPI cloud relay.
- Offer all desktop users 10 GB of free cloud-coordinated deduplication credits.
- Introduce $4.99 / 100 GB Pay-As-You-Go credit packs for high-volume creators.

Phase 3: Enterprise Team Tier Expansion

- Launch Team Starter ($29/mo) and Business Premium ($99/mo) with centralized multi-device pairing and fleet-wide storage reporting.